

# NATIONAL PHARMACOPOEIAL ANALYTICAL LABORATORY

Accredited by NABL (ISO/IEC 17025:2017) & Recognized by Ministry of AYUSH, Govt. of India  
Central Drug Testing Facility, Sector-62, Institutional Area, New Delhi - 110001

NABL CERT: TC-  
8492

## CERTIFICATE OF ANALYSIS (CoA)

**Sample Name:** Ashwagandha Dried Root Churna **Report Number:** NPAL/AYUSH/2026/0842  
**Botanical Species:** *Withania somnifera* (L.) Dunal **Batch Quantity:** 10.0 kg  
**Monograph Reference:** Ayurvedic Pharmacopoeia of India (API Vol. I) **Date of Analysis:** 04-Oct-2026

### 1. Quantitative Physicochemical & Active Marker Analysis

| Parameter                           | Test Method            | Pharmacopoeial Specification | Observed Value       | Result   |
|-------------------------------------|------------------------|------------------------------|----------------------|----------|
| <b>Botanical Purity &amp; Assay</b> | HPTLC Densitometry     | ≥ 95.0% Botanical Identity   | <b>Purity: 98.4%</b> | CONFORMS |
| Withanolide Content (A + D)         | HPLC-UV (227 nm)       | ≥ 0.50% w/w                  | 0.82% w/w            | CONFORMS |
| Loss on Drying (LOD)                | Gravimetric (105°C)    | ≤ 10.0% w/w                  | 6.4% w/w             | CONFORMS |
| Total Ash Content                   | Muffle Furnace (550°C) | ≤ 7.0% w/w                   | 4.8% w/w             | CONFORMS |
| Acid-Insoluble Ash                  | API Part-I Appendix 2  | ≤ 1.0% w/w                   | 0.42% w/w            | CONFORMS |

### 2. Contaminant, Heavy Metals & Safety Profiling

| Safety Parameter                     | Analytical Method             | Permissible AYUSH Limit | Detected Value              | Status    |
|--------------------------------------|-------------------------------|-------------------------|-----------------------------|-----------|
| <b>Foreign Organic Contamination</b> | Visual / Microscopy (USP 561) | Nil (0.00)              | <b>Contamination: 0</b>     | COMPLIANT |
| Lead (Pb)                            | ICP-MS (EPA 6020B)            | ≤ 10.0 ppm              | 0.45 ppm                    | COMPLIANT |
| Arsenic (As)                         | ICP-MS (EPA 6020B)            | ≤ 3.0 ppm               | 0.12 ppm                    | COMPLIANT |
| Aflatoxins (B1, B2, G1, G2)          | LC-MS/MS                      | ≤ 5.0 ppb               | Not Detected (<0.5 ppb)     | COMPLIANT |
| Total Microbial Plate Count          | Pour Plate Method             | ≤ 10 <sup>5</sup> CFU/g | 3.2 x 10 <sup>2</sup> CFU/g | COMPLIANT |

✓ FINAL PHARMACOPOEIAL OPINION: APPROVED FOR MEDICINAL EXTRACTION

The submitted raw herb batch complies with all analytical monographs laid down in the **Ayurvedic Pharmacopoeia of India (API)** and WHO Good Agricultural and Collection Practices (GACP). The raw biomass is certified **APPROVED** for commercial capsule and extract manufacturing.

**Dr. P. K. Sharma**  
Lead Analytical Chemist

**Dr. V. N. Raghunathan**  
Quality Assurance Director

**Official Seal & Signature**  
Govt. Approved Testing Lab